CHAPTER 2

MOTION IN A STRAIGHT LINE

CHAPTER ESSENCE AT A GLANCE:
Moving along a straight line is as easy as ABC! Distance is how far you went; displacement is the straight shortcut. The area under a velocity-time graph gives total displacement, and its slope gives acceleration. Three simple equations solve 95% of all board questions!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Distance vs Displacement:** Distance is the total path length traveled (scalar, always ≥ 0). Displacement is the shortest straight line from start to finish (vector, can be positive, negative, or zero).

• **Graph Interpretation:** Slope of Position-Time (x-t) graph = Velocity (v). Slope of Velocity-Time (v-t) graph = Acceleration (a). Area under v-t graph = Displacement (s).

• **Kinematic Equations:** Valid ONLY for uniform (constant) acceleration: (1) v = u + at, (2) s = ut + ½at^2, (3) v^2 - u^2 = 2as, (4) s nth = u + a divided by 2 (2n - 1).

• **Sign Convention in Free Fall:** Take upward as positive (+). Acceleration due to gravity is always strictly downwards: a = -g = -9.8 m/s^2 (or -10 m/s^2). At maximum height, velocity is momentarily ZERO (v = 0).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$v = u + at \quad \text{and} \quad s = ut + \tfrac{1}{2}at^2$$

First and Second Kinematic Equations

$$v^2 = u^2 + 2as \quad \text{and} \quad s_n = u + \tfrac{a}{2}(2n - 1)$$

Third Equation and Distance in nth Second

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}, \quad a = \frac{dv}{dt} = v\frac{dv}{dx}$$

Average Velocity and Acceleration Derivatives

$$d_{\text{stop}} = \frac{u^2}{2a}$$

Stopping distance under uniform retardation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Always watch sign conventions! In free fall, if upward is taken as positive, then acceleration due to gravity g is NEGATIVE (-9.8 m/s^2).

• **Alert:** When a question says 'starts from rest', immediately set u = 0. When it says 'comes to rest or stops', immediately set v = 0!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Differentiate between distance and displacement with two key differences. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Comparison Points]** 1. Distance is the total actual length of the path traversed by a body (scalar quantity, always ≥ 0). [1 Mark]

• 2. Displacement is the shortest straight-line vector from the initial to the final position (vector quantity, can be positive, negative, or zero). [1 Mark]

• **Key Exam Result:** Distance (scalar, path length) vs Displacement (vector, shortest distance)

2. Draw position-time (x-t) graphs for: (a) A body at rest (b) A body moving with uniform velocity. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Body at Rest]** (a) A horizontal straight line parallel to the time axis (slope = 0, indicating v = dx divided by dt = 0). [1 Mark]

• **[Step 2: Uniform Velocity]** (b) An inclined straight line passing through origin or intercept with constant non-zero slope (constant velocity v). [1 Mark]

• **Key Exam Result:** Rest: horizontal line; Uniform v: straight inclined line

3. Derive the kinematic equation v = u + a t for uniformly accelerated motion using a velocity-time (v-t) graph. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Graph Setup]** Let an object move with initial velocity u at t = 0 and accelerate uniformly to final velocity v at time t. The v-t graph is a straight line AB from (0, u) to (t, v). [1 Mark]

• **[Step 2: Slope Formulation]** Acceleration a = slope of v-t line AB = BC divided by AC = (v - u) divided by (t - 0). [1 Mark]

• **[Step 3: Algebraic Rearrangement]** a = (v - u) divided by t => a t = v - u => v = u + a t. [1 Mark]

• **Key Exam Result:** v = u + a t

4. Derive the displacement relation $s = u t + ½ a t^2$ from the velocity-time (v-t) graph. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Area Under Graph]** The total displacement s is equal to the area under the v-t graph between time t = 0 and time t (Trapezium OABD). [1 Mark]

• **[Step 2: Geometric Splitting]** Area = Area of Rectangle OACD + Area of Triangle ACB = (OA x OD) + (half (0.5) x AC x BC). Here OA = u, OD = AC = t, and BC = v - u = a t. [1 Mark]

• **[Step 3: Final Substitution]** s = (u x t) + half (0.5) x t x (a t) = u t + ½ a t^2. [1 Mark]

• **Key Exam Result:** $s = u t + ½ a t^2$

5. Using calculus method (integration), derive the relation $v^2 = u^2 + 2 a s$ for a body moving with uniform acceleration. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definition of Acceleration]** a = dv divided by dt = (dv divided by ds) x (ds divided by dt) = v (dv divided by ds). [1 Mark]

• **[Step 2: Variable Separation]** a ds = v dv. Integrating both sides within limits s: 0 -> s and v: u -> v: a $\int_0^s$ ds = $\int_u^v$ v dv. [1 Mark]

• **[Step 3: Integration Evaluation]** a [s] 0^s = [v^2 divided by 2] u^v ⇒ a s = (v^2 - u^2) divided by 2 ⇒ 2 a s = v^2 - u^2 ⇒ v^2 = u^2 + 2 a s. [1 Mark]

• **Key Exam Result:** $v^2 = u^2 + 2 a s$

6. A car moving along a straight road with a speed of 126 km/h is brought to a stop within a distance of 200 m. Find the retardation of the car and the time taken to stop. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Conversion of Units]** Initial velocity u = 126 km/h = 126 x (5 divided by 18) = 35 m/s. Final velocity v = 0, distance s = 200 m. [1 Mark]

• **[Step 2: Retardation Calculation]** From $v^2 = u^2 + 2as$: $0 = 35^2 + 2a(200) \Rightarrow 400a = -1225 \Rightarrow a = -3.06$ m/s^2. Retardation = 3.06 m/s^2. [1 Mark]

• **[Step 3: Time Taken]** From v = u + a t: 0 = 35 - 3.06 t => t = 35 divided by 3.06 = 11.44 s. [1 Mark]

• **Key Exam Result:** Retardation = 3.06 m/s^2, Time = 11.44 seconds

7. A ball is thrown vertically upwards with a velocity of 20 m/s from the top of a multi-storey building 25 m high. Find: (a) The maximum height reached above the building (b) The total time taken before hitting the ground. (Take g = 10 m/s^2) [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Maximum Height]** At peak: v = 0, u = +20 m/s, a = -10 m/s^2. $0 = 20^2 - 2(10)h \Rightarrow 20h = 400 \Rightarrow h = 20$ m. Max height above ground = 25 + 20 = 45 m. [1.5 Marks]

• **[Step 2: Total Time to Hit Ground]** Net displacement from top to ground: s = -25 m. $s = ut + \frac{1}{2}at^2 \Rightarrow -25 = 20t - 5t^2 \Rightarrow 5t^2 - 20t - 25 = 0 \Rightarrow t^2 - 4t - 5 = 0$. Factoring: (t - 5)(t + 1) = 0 => t = 5 s (rejecting t = -1 s). [1.5 Marks]

• **Key Exam Result:** (a) Height = 20 m above roof (45 m from ground), (b) Time = 5 seconds